

AEEE 2026 — Official Syllabus

Amrita Entrance Examination – Engineering | Amrita Vishwa Vidyapeetham | Computer Based Test

■ **Overview**

AEEE 2026 is a national-level CBT for B.Tech admissions to Amrita Vishwa Vidyapeetham. Conducted in two phases (typically February and May 2026), with applications opening October–November 2025. Held across 100+ cities in English. Admission also open via JEE Main percentile — candidates can apply through both routes. Application fee: ■1,300 (Phase 1); ■600 additional for Phase 2 (optional).

■ **Exam Pattern**

Subject	Questions	Marks (×3)	Notes
Mathematics	40	120	Highest weightage
Physics	25	75	
Chemistry	20	60	
Quantitative Aptitude	10	30	
English	5	15	
Total	100	300	+3 correct / –1 wrong

Duration: 150 minutes (2.5 hours) · Mode: CBT at 100+ cities · Language: English only

■ **Eligibility Criteria**

Criterion	Requirement
Age	Date of birth on or after July 1, 2005
Marks	Minimum 60% aggregate in Mathematics, Physics & Chemistry; at least 50% individually in each of PCM
Qualifying Exam	10+2 from CBSE, CISCE, State Boards, or equivalent (Pre-University, GCE A-Level, Cambridge, IB Diploma, NIOS with 5 subjects)
JEE Route	Admission also available via JEE Main percentile — no additional fee if already registered for AEEE 2026
Phases	Phase 2 is optional. Attend Phase 1 to test skills, improve and appear in Phase 2 for ■600 extra

■ **MATHEMATICS — 40 Questions · 120 Marks**

Algebra	Complex numbers, Quadratic equations, Sequences & series (AP, GP), Permutations & Combinations, Binomial theorem, Matrices & Determinants, Sets/Relations/Functions
Trigonometry	Trigonometric functions, Identities, Equations, Inverse trig functions, Heights & distances
Coordinate Geometry (2D)	Straight lines, Circles, Parabola, Ellipse, Hyperbola — standard forms, tangents, normals
Coordinate Geometry (3D)	Direction cosines, Lines in space, Planes, Skew lines, Shortest distance
Differential Calculus	Limits, Continuity, Derivatives (all types), Maxima/Minima, Rolle's theorem, MVT
Integral Calculus	Indefinite & definite integrals, Integration methods, Area under curves
Differential Equations	Order & degree, Variables separable, Homogeneous, Linear first order
Vectors	Addition, Dot & cross products, Scalar triple product
Probability & Statistics	Probability rules, Bayes' theorem, Random variables, Binomial & Normal distributions, SD, Variance
Linear Programming	Graphical method, Feasible regions, Optimal solutions

■ **PHYSICS — 25 Questions · 75 Marks**

Mechanics	Newton's laws, Work-Energy theorem, Rotational motion, Gravitation, Fluid mechanics, Elasticity
Oscillations & Waves	SHM, Spring-mass, Pendulums, Standing waves, Beats, Doppler effect

Heat & Thermodynamics	Kinetic theory, Specific heat, Heat transfer, First & second laws of thermodynamics, Carnot engine
Electrostatics	Coulomb's law, Electric field, Gauss's law, Capacitance, Dielectrics
Current Electricity	Ohm's law, Kirchhoff's laws, Potentiometer, Wheatstone bridge, Resistivity
Magnetism	Biot-Savart's law, Ampere's law, Moving coil galvanometer, Magnetic dipole
Electromagnetic Induction & AC	Faraday's law, Lenz's law, Self/Mutual inductance, LCR circuits, AC generator
Optics	Reflection, Refraction, TIR, Lenses, Young's double slit, Diffraction, Polarization
Modern Physics	Photoelectric effect, de Broglie, Bohr's model, Radioactivity, Nuclear fission & fusion
Electronic Devices	P-N junction, Transistors, Logic gates

■ CHEMISTRY — 20 Questions · 60 Marks

Physical Chemistry	Atomic structure, Chemical bonding (VSEPR/VBT/MOT), States of matter, Thermodynamics, Chemical kinetics, Electrochemistry
Solutions & Equilibrium	Raoult's law, Colligative properties, van't Hoff factor, KP & KC, Le Chatelier, pH, Buffer, Ksp
Inorganic Chemistry	Periodic trends, s/p/d/f block elements (groups 1-18), Coordination compounds (IUPAC, VBT, CFT, isomerism)
Organic Chemistry — Basics	Hybridization, Functional groups, IUPAC nomenclature, Inductive/Resonance/Hyperconjugation effects
Organic Chemistry — Reactions	Alkanes, Alkenes, Alkynes, Benzene, Haloalkanes, Alcohols, Phenols, Aldehydes, Ketones, Carboxylic acids, Amines
Biomolecules & Polymers	Carbohydrates, Proteins, Nucleic acids (DNA/RNA), Vitamins, Natural & synthetic polymers
Environmental Chemistry	Air/Water pollution, Greenhouse effect, Ozone depletion, Green chemistry

■ QUANTITATIVE APTITUDE — 10 Questions · 30 Marks

Number System	HCF, LCM, Fractions, Decimals, Powers & roots
Arithmetic	Percentage, Profit & loss, Simple & compound interest, Ratio & proportion, Time-speed-distance
Algebra	Equations, Inequalities, Sequences
Data Interpretation	Tables, Pie charts, Bar graphs, Line graphs
Logical Reasoning	Series, Analogies, Coding-decoding, Blood relations, Direction sense

■ ENGLISH — 5 Questions · 15 Marks

Grammar	Tense, Voice, Parts of speech, Sentence correction
Vocabulary	Synonyms, Antonyms, Idioms, One-word substitution
Reading Comprehension	Short passages — main idea, inference, vocabulary in context